

An Intelligent Car-following Model Based on Multi-step Deep Reinforcement Learning

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Abstract—With the development of vehicular networking technology, the development of car-following model that serves for intelligent and/or autonomous driving has been an interested topic. Deep reinforcement learning (DRL) based models are attracting considerable attentions. However, due to the complex traffic environment, development of an efficient car-following model is a challenge issue in terms of extended training times, low sample utilization rates, and poor stability. To deal with these issues, this work proposes a reward processing mechanism that considers the cumulative impact of multi-step reward. The proposed DRL algorithm shows expedited learning speed, enhanced sample utilization and effectively resolved the challenges encountered during training. To evaluate the proposed car-following model, SUMO, an open source for micro, multi-modal traffic simulation is used to construct a realistic driving environment. Finally, the results with respect to several driving scenario validate the model, and the comparison results to the widely discussed intelligent driver model (IDM) model show the advancements of the proposed model.

Keywords—*car-following model, deep reinforcement learning, multi-step DQN, validation*

I. INTRODUCTION

Car following has always been an integral part of autonomous driving technology, and an intelligent car-following model is continuously being optimized and investigated.

In 1958, Gipps proposed the first car-following model based on vehicle spacing and speed differentials, known as the Gipps model [1]. It considers the expected speed of vehicles and safe following distances, providing a more realistic description of car-following behavior. This enables the simulation of car-following behavior at different speeds, making it suitable for various traffic flow conditions. In 1981, Bando and others introduced a macroscopic car-following model based on differential equations, known as the Bando model [2]. The Bando model considers interactions between vehicles and wave phenomena, effectively describing the behavior of congested traffic flow. In 1995, Treiber and Helbing proposed a microscopic simulation-based car-following model, known as the Treiber-Helbing model [3]. The Treiber-Helbing model combines microscopic and macroscopic features to effectively simulate the dynamic behavior of traffic flow. In 2000, Krauß and others introduced

an acceleration-based car-following model known as the Krauß model [4]. The Krauß model considers the mutual influence between vehicles and the effect of acceleration, providing a more accurate description of car-following behavior. However, classical car-following models are often based on simplified assumptions and parameters, which are unable to fully describe complex traffic flow behavior. This limitation has led to the gradual development of modern car-following models.

In 2000, Treiber and Helbing combined the classic car-following model with intelligent control algorithms to propose the IDM model [5]. This model can accurately simulate the interaction behavior between vehicles, thereby enhancing the efficiency and safety of traffic flow. With the advent of the big data era, deep learning has been widely applied in car-following models because it can learn complex patterns and features from large-scale data [6]. The problem of car-following is a complex decision-making issue. Deep reinforcement learning (DRL) can learn the optimal strategy in complex environments, adapt to different road conditions and traffic situations. Car-following models build with DRL have become a research hotspot [7].

In recent years, some studies have compared two models of car-following behavior: one based on DRL and the other based on classical car-following models in the same simulation environment. It was found that the car-following model based on deep reinforcement learning performs equally or even better. In the work [8], a self-decision-making method based on Q-learning is proposed, establishing a Markov decision model based on car-following behavior. In the work [9], an ecological driving strategy based on Twin Delayed Deep Deterministic Policy Gradient (TD3) agents was proposed. This strategy involved creating a probabilistic traffic simulation model with traffic lights and multiple preceding vehicles. The work [10] proposes an automatic driving car safety speed control method based on reinforcement learning was proposed. It constructs a mixed driving environment with autonomous driving cars and human-driven cars. In the work [11], a city driving strategy learning model for car-following based on DRL was proposed. The work [12] focuses on the reinforcement learning-based vehicle tracking, and shows that the proposed model can effectively manage safety-critical situations.

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In the above-mentioned study, although the car-following model established using DRL algorithms has better performance. However, in today's complex and ever-changing autonomous driving environment, more diverse and stringent requirements have been proposed. Car-following models based on DRL need to exhibit higher performance, stability, and adaptability. In order to address this issue, two targeted solutions have been developed. One solution is aimed at meeting the diverse requirements of autonomous driving environments. This solution considers that the model has better adaptability and performance in a complex and variable car-following environment. The work [13] introduces an anti-interference car-following strategy to mitigate external disturbances from traffic oscillations ahead and internal disturbances in the vehicle control system, enhancing the model's resilience to interference. The work [14] introduces a two-dimensional integrated control strategy for autonomous driving cars based on DRL. This strategy effectively manages the two-dimensional following process of autonomous driving cars by enhancing longitudinal stability control performance and lateral precise path tracking performance. It enables safe driving on roads with varying curvatures.

Another approach is to meet the more stringent requirements in the autonomous driving environment by enhancing the structure and design of the model, improving its performance and adaptability. Most reinforcement learning car-following models only perceive the state data of the preceding vehicle. The work [15] introduces a bilateral car-following strategy based on DRL that integrates the information from both the preceding and following vehicles into the state and reward functions. The work [16] proposes a hybrid strategy, combining classic car-following strategies with DRL strategies. This approach combines the advantages of knowledge-driven and data-driven models. In the work [17], a stacked denoising autoencoder (SDAE) is utilized to enhance the deep deterministic policy gradient (DDPG) car-following model by introducing a data-driven reward representation function.

Although various methods have been developed to improve the performance and adaptability of DRL car-following models, there is still a lack of techniques to enhance these models from the perspective of sample data. This study enhances the reward processing mechanism of the model by implementing multi-step reward accumulation, with the goal of improving the performance and stability of DRL models.

The main contributions of this work are as follows:

- The traditional DQN algorithm only considers the current reward and the Q-value of the next state at each time step. In complex and dynamic autonomous driving environments, longer training times are required, which can result in issues such as low sample utilization and model instability. By accumulating multi-step rewards to enhance the reward processing mechanism of the model, the stability and effectiveness of training are improved.
- A new reward mechanism has been designed to enhance the efficiency, safety, and comfort of car-following behavior. Under this reward mechanism, DRL can enhance car-following behavior, making it more effective and realistic.

In order to further demonstrate the DRL car-following model considering multi-step rewards and the ordinary DRL car-following model, we built three simulation scenarios in

SUMO. These scenarios include real-life situations where preceding vehicles accelerate, drive at a constant speed, and decelerate. We evaluated the performance of the two models in terms of training and testing. The simulation results show that the improved model has better performance.

The organization of this work is as follows. The problem statement is shown in Section 2. Section 3 introduces the improved DRL based car-following model. Section 4 shows the validation results and Section 5 concludes the work.

II. PROBLEM STATEMENT

Reinforcement learning focuses on how an intelligent agent maximizes the rewards it can obtain in a complex and uncertain environment. This is achieved by perceiving the state of the environment and using the rewards of actions to guide better actions, ultimately leading to the attainment of maximum profit. Due to the trial-and-error learning and delayed reward characteristics of reinforcement learning, the intelligent agent must continuously interact with the environment to acquire the optimal strategy. In some straightforward application scenarios, the intelligent agent can quickly determine the optimal strategy through interaction with the environment. However, in the context of autonomous driving, which involves a large amount of data, reinforcement learning training takes a long time to achieve the optimal behavior for car-following.

The work of this paper focuses on the DRL car-following model. The environment is as illustrates by Fig. 1, which depicts a classic car-following scenario.

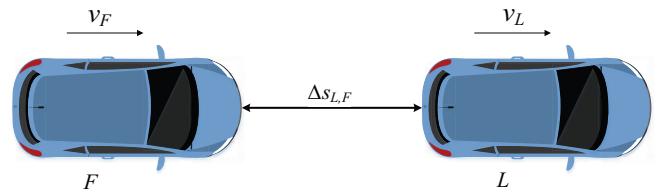


Fig. 1 Car-following scenario

Define the leading vehicle as L , with position d_L , velocity v_L and acceleration a_L ; the following vehicle is defined as F , with position d_F , velocity v_F , and acceleration a_F , $\Delta s_{L,F}$ represents the distance between the leading vehicle and the following vehicle, $\Delta v_{L,F}$ represents the speed difference between the leading vehicle and the following vehicle.

The aim is to enhance the training efficiency and effectiveness of DRL car-following models in complex environments. Data plays a crucial role in reinforcement learning models, as these models learn optimal behavioral strategies through interaction with the environment. This interaction process heavily relies on data, which helps the model understand the environment's state and reward signals, guiding the model to make correct decisions in various situations. Starting from the data and combining it with the method of multi-step reward accumulation to enhance the way the model processes rewards, the reward handling mechanism is not solely reliant on the reward acquired by the agent for taking action in the current state; it also considers the rewards obtained in the subsequent two steps. This is done to increase the utilization of sample data, thereby enhancing the training effectiveness and efficiency of DRL models for car-following.

Fig.2 shows the improved DRL agent and environment interaction architecture. The figure illustrates the process of improving the interaction between the DQN agent and the environment to acquire the state (s_t), action (a_t), reward (r_t), and next state (s_{t+1}) utilized during training. Firstly, the intelligent system observes the current environment and obtains s_t . The value ($Q(s_t, a_t)$) of all actions taken by s_t is obtained through the Q network. a_t taken in s_t is determined by a random policy function, and r_t for taking a_t in s_t is obtained through a reward function. The DQN agent implements a_t and obtains s_{t+1} . The processes obtained by r_{t+1} and r_{t+2} are the same as r_t . Finally, The function accumulates r_t , r_{t+1} , and r_{t+2} to obtain R_t . R_t contains information about r_t , r_{t+1} , and r_{t+2} , which can better assist in training models compared to r_t .

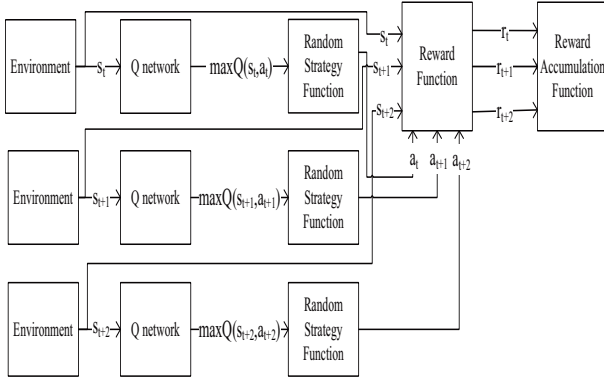


Fig. 2 Improved version of DRL agent and environment Interaction Architecture

III. DRL-BASED CAR-FOLLOWING MODEL

A. Markov Decision Process

Car-following behavior is one of basic micro-driving behaviors, and it describes the interaction between two adjacent vehicles driving on a single lane road with overtaking restrictions. During the car-following process, the following vehicle will make decisions based on the leading vehicle and its own state to maintain a safe following distance. Due to the Markovian nature of car-following behavior, where the transition to future states depends only on the current state and decision, modeling the car-following problem as a Markov decision process allows for the optimization of car-following efficiency using reinforcement learning methods to improve driving safety and efficiency.

Define the states, actions, rewards, and discount factor of the car-following behavior based on the MDP quintuple. The agent is the following vehicle controlled by DRL, and the environment is simulated in SUMO. Due to the complexity of the task, the transition probabilities of states are unknown.

State describes the situation in which an intelligent agent finds itself at a certain moment. For the considered problem, define the state as: $S = \{v_F(t), \Delta s_{L,F}(t), \Delta v_{L,F}(t)\}$.

Action describes the behavior that an intelligent agent can perform in a specific state. In this work, consider the desired acceleration of the vehicle as the action.

In reinforcement learning framework, the goal of the agent is to maximize long-term cumulative rewards, hence the setting of rewards is crucial. For the considered car-following modeling problem, the defined rewards include three categories: safety, efficiency, and comfort.

1) Safety

For the safety of the reward function, we base it on Time to Collision (TTC). TTC represents the remaining time before the lead vehicle and the following vehicle collide, with the calculation formula being equation (1).

$$TTC(t) = \frac{\Delta s_{L,F}(t)}{\Delta v_{L,F}(t)} \quad (1)$$

When the TTC is less than the threshold we set, we penalize it. We assume that the leading vehicle and the following vehicle will not collide, otherwise, the reward is set as 0. The function is defined as equation (2).

$$f_{safety}(TTC) = \begin{cases} \log\left(\frac{TTC}{2}\right), & 0 \leq TTC \leq 2 \\ 0, & \text{else} \end{cases} \quad (2)$$

2) Efficiency

For the efficiency of the reward function H , which represents the time interval when the leader and the follower reach the same location. The formulation is as shown by equation (3).

$$H(t) = \frac{\Delta s_{L,F}(t)}{v_F(t)} + \frac{\Delta s_{L,F}(t)}{\Delta v_{L,F}(t) + v_F(t)} \quad (3)$$

Expect the time difference between the leading vehicle and the following vehicle passing through the same point to be no more than 5 seconds. The function is defined as equation (4):

$$f_{eff}(H) = \begin{cases} -|H - 5|, & H \geq 5 \\ 0, & \text{else} \end{cases} \quad (4)$$

3) Comfort

For the comfort of the reward function, we base it on *jerk*, which represents the rate of change in acceleration. The calculation formula is given by equation (5).

$$jerk(t) = \frac{a_F(t) - a_F(t-1)}{\Delta t} \quad (5)$$

The higher the *jerk*, the heavier the penalty for it, as defined by the equation (6).

$$f_{com}(jerk) = -\frac{jerk^2}{100} \quad (6)$$

Therefore, the reward function is set as the following equation (7):

$$r = \omega_s \cdot f_{safety} + \omega_e \cdot f_{eff} + \omega_c \cdot f_{com} \quad (7)$$

Finally, adjusting the proportion of safety, efficiency, and comfort in the overall reward function by the weighting factors ω_s , ω_e , and ω_c , respectively.

Mark the reward obtained at time t , $t + 1$, and $t + 2$ as r_t , r_{t+1} , and r_{t+2} , respectively, and the multi-step merged reward at time t as R_t . The relationship among them is shown in equation (8):

$$R_t = r_t + r_{t+1} + r_{t+2} \quad (8)$$

B. Deep Reinforcement Learning Algorithm

In this problem, we consider the actions of the intelligent agent as discrete, while the states of the intelligent agent are continuous. We have chosen the Deep Q-Network (DQN) algorithm to solve this problem, which combines Q-learning with deep neural networks. The deep neural network enables this algorithm to handle situations where the states are continuous. The DQN structure consists of an experience replay pool, a Q network, a target Q network, and a greedy policy function. The experience replay pool stores historical data of the interactions between the agent and the environment. During model training, a batch of data can be randomly sampled from the dataset, without being limited to using only the most recent data., a sample $\langle s_t, a_t, R_t, s_{t+1} \rangle$ involving the interaction with the environment is stored in experience replay pool. The Q network is utilized to estimate the Q value of taking actions in the current state, while the target Q network is employed to calculate the target Q value. During model training, the weights of the Q network are updated at each time step. Additionally, every five time steps, the weights of the Q network are copied to the target Q network. The greedy policy uses an ϵ -greedy random exploration strategy, where with a probability of ϵ , exploration behavior is adopted, and with a probability of $1 - \epsilon$, greedy behavior is adopted.

IV. EMPIRICAL STUDY

A. Simulation Environment

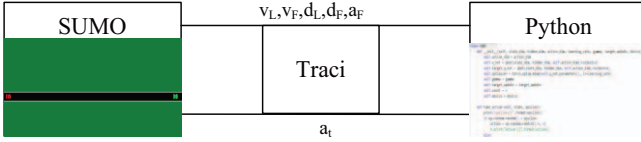


Fig. 3 Structure of the constructed simulation platform

To validate the improved DRL car following model, we employed a joint simulation approach using Python and SUMO Fig.3. We utilized the Traci library to establish the connection between Python and SUMO. The DQN algorithm used in this study is implemented in Python, while the three simulation scenarios designed are conducted in the SUMO environment. Through the Traci interface, Python transmits the desired acceleration of the following vehicle to SUMO and receives the positions, velocities, and accelerations of the leading and following vehicles transmitted by SUMO.

We utilize SUMO to create a simulation environment, which includes roads, following vehicles, and leading vehicles. The road is a 1000m straight road, with the initial speed v_0 of the following vehicle being 1m/s, the initial position d_0 being 20m, and the initial acceleration a_0 being 1m/s². It is controlled by a DRL intelligent agent, while the leading vehicle is controlled by SUMO. The simulation environment is utilized to simulate the status of the vehicle under different conditions: acceleration mode, constant speed mode, and deceleration mode when the front vehicle is in those states.

The specific settings are as follows:

- Acceleration mode: In acceleration mode, we set the initial velocity of the leading vehicle to be $v_0 = 1\text{m/s}$, the initial position to be $d_0 = 22\text{m}$, and the initial acceleration to be $a_0 = 1\text{m/s}^2$.
- Constant speed mode: In the constant speed mode, we set the initial speed of the leading vehicle to be $v_0 =$

1m/s, the initial position to be $d_0 = 22\text{m}$, and the initial acceleration to be $a_0 = 0\text{m/s}^2$.

- Deceleration mode: In deceleration mode, we set the initial speed of the leading vehicle as $v_0 = 1\text{m/s}$, initial position as $d_0 = 22\text{m}$, and initial acceleration as $a_0 = -1\text{m/s}^2$.

B. Training Process

We set the training period to 200 episodes, batch size to 100, and the target network update steps to 5. The probability of exploration starts at 0.9, decreases by 0.01 each time, and stops at a probability of 0.1.

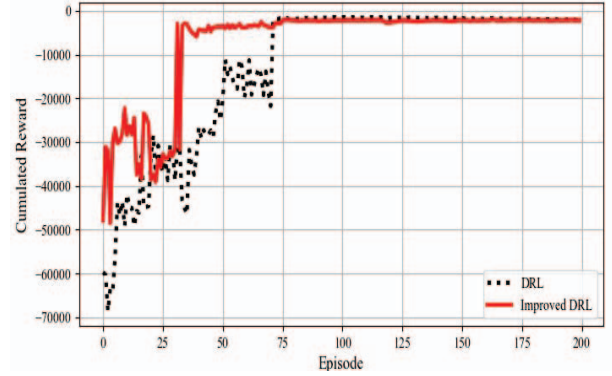


Fig. 4 Training process: The trend of changes in cumulative rewards for DRL and improved DRL.

Fig.4 illustrates the trend of cumulative rewards during the model training process for DRL and improved DRL. It is clear from the figure that the improved DRL tends to stabilize after around 30 training episodes, while the standard DRL tends to stabilize after around 70 training episodes. Therefore, it can be seen that the improved DRL significantly improves training efficiency.

C. Evaluation Process

1) Scenario 1: Acceleration Mode

Fig.5 illustrates the impact of the vehicle under the control of the IDM model, DRL model, and Improved DRL model when the leading vehicle is in acceleration driving mode. The first figure describes the distance covered by a vehicle under the control of three different models. The improved DRL model has the best control effect, with a control distance of less than 100m, while the IDM model has the worst control effect, with a control distance of less than 400m. The second figure illustrates the acceleration of the vehicle under three different model controls. The IDM model has the best control effect, while the improved DRL model has a moderate control effect. The third figure illustrates the speed of the vehicle under three different model controls. The IDM model has the best control effect, while the improved DRL model has moderate control effect. The improved DRL model can maintain optimal tracking performance and provide moderate comfort in an accelerated driving scenario.

2) Scenario 2: Constant Speed Mode

Fig.6 illustrates the impact of the vehicle under the control of the IDM model, DRL model, and Improved DRL model when the leading vehicle is in constant speed driving mode. The first figure describes the distance covered by a vehicle under the control of three different models. The improved DRL model has the best control effect, with a control distance of maintain around 1m, while the IDM model

has the worst control effect, with a control distance of maintain around $1m$. The second figure illustrates the acceleration of the vehicle under three different model controls. The IDM model has the best control effect, while the improved DRL model has a moderate control effect. The third figure illustrates the speed of the vehicle under three different model controls. The control effect of IDM is consistent with the acceleration phenomenon, while the improved DRL model has moderate control effect. The improved DRL model can maintain optimal tracking performance and provide moderate comfort in a constant speed driving scenario.

3) Scenario 3: Deceleration Mode

Fig.7 illustrates the impact of the vehicle under the control of the IDM model, DRL model, and Improved DRL model when the leading vehicle is in deceleration driving mode. The first figure describes the distance covered by a vehicle under the control of three different models. The improved DRL model has the best control effect, with a

control distance of less than $5m$, while the IDM model has the worst control effect, with a control distance of less than $9m$. The second figure illustrates the acceleration of the vehicle under three different model controls. The DRL model and the improved DRL model demonstrate good performance, whereas the control effectiveness of the Intelligent Driver Model (IDM) model is very poor. The third figure illustrates the speed of the vehicle under three different model controls. The IDM model has the worst control effect, while the improved DRL model has moderate control effect. The improved DRL model can maintain optimal tracking performance and provide moderate comfort in a deceleration driving scenario.

In order to verify the effectiveness of the improved DRL model, we envisioned three scenarios and established a simulation environment using SUMO. Among them, we compared the improved DRL model with the DRL model and the classic car-following model, the Intelligent Driver Model (IDM), in terms of distance, acceleration, and speed. After simulation results, we can see that compared with the other two models, the improved DRL model has the best

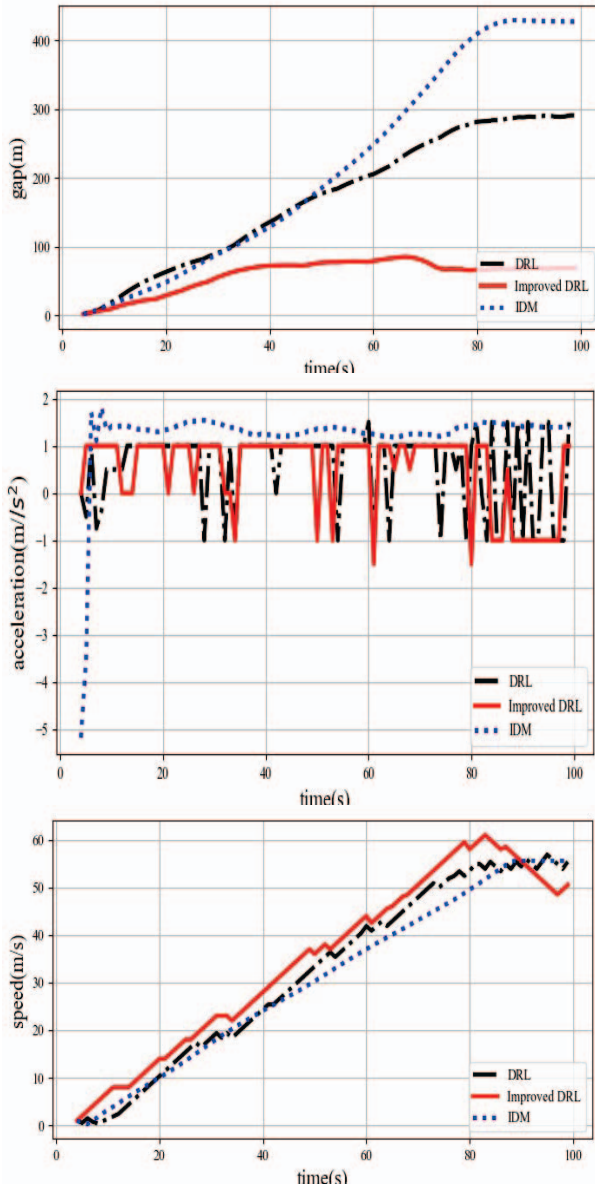


Fig. 5 Test results: The distance, acceleration, and speed of IDM, DRL, and improved DRL

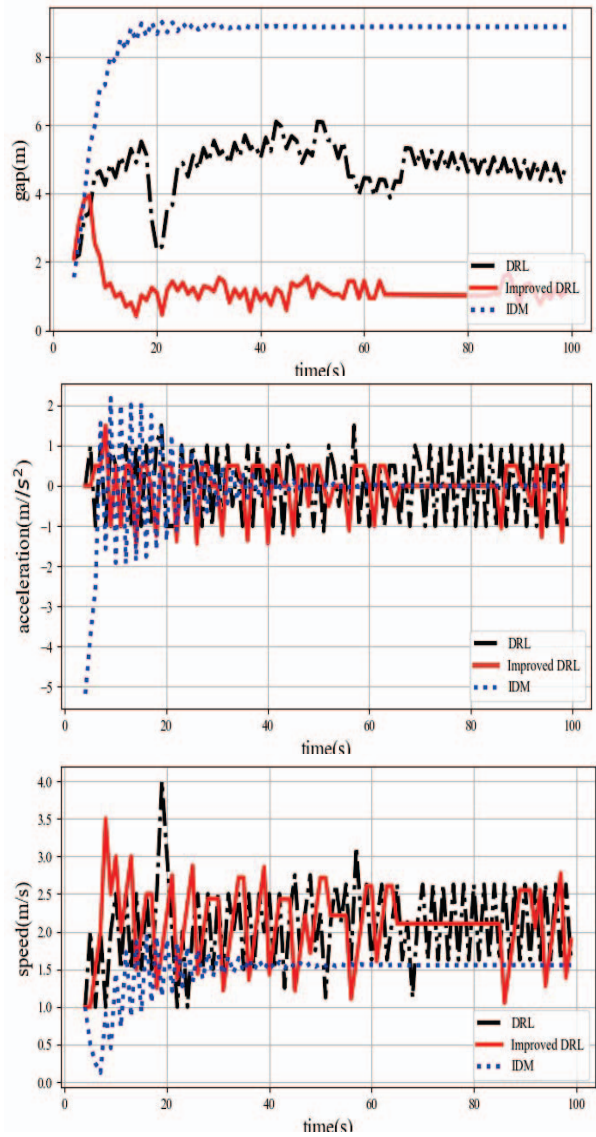


Fig. 6 Test results: The distance, acceleration, and speed of IDM, DRL, and improved DRL

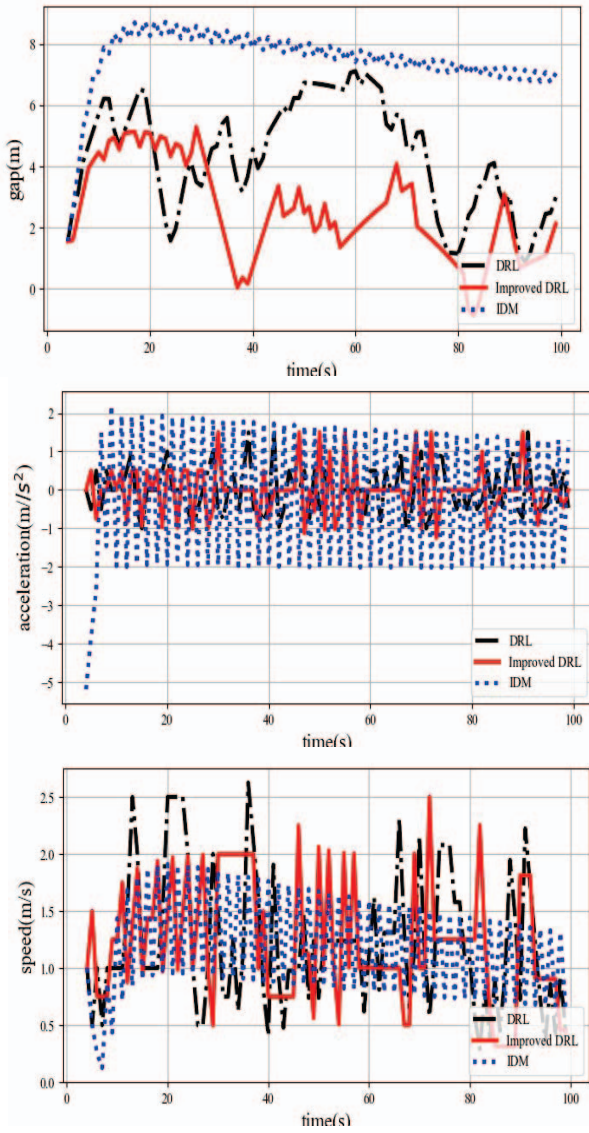


Fig. 7 Test results: the distance, acceleration, and speed of IDM, DRL, and improved DRL

performance in terms of following distance, moderate acceleration, and speed. From this, it can be seen that the improved DRL model demonstrates a certain level of effectiveness.

V. CONCLUSION

In this work, a multi-step reward accumulation method is proposed to develop an improved DRL based car-following model. By comparison with the methods of using traditional DRL algorithm and IDM, the proposed improved DRL algorithm shows significant advantages in both training effectiveness and model predictive efficiency with respect to different operating scenarios. When the weighting factors are same in the reward function, the results show that the proposed model guarantees improved following performance, however, the comfortability is decreased. Regarding the multi-objective optimization problem, how to find the Pareto solution is significant, moreover, the proposed model is only evaluated with respect to car-following behavior over time. In the future work, the influence factors from lane changing

should be considered in the car-following model development.

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